

Post-quantum migration planning for manufacturing OT and IT systems to secure long-lived industrial data.

Manufacturing × Post-quantum cryptography migration × Post-quantum cryptography · OS007 v2 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
71 is the world moving	71 can we play, can we win	MEDIUM 0 named in the evidence	€3.4bn partial evidence · per year	LATER research and standards activi...	L0 15 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity helps manufacturers plan and execute migration to post-quantum cryptography (PQC) for their OT and IT systems, protecting long-lived industrial data from future quantum threats. It targets manufacturing firms with complex, long-lived systems and sensitive data, driven by NIST standards finalization and looming migration deadlines. The time is now because standards are set, but enterprise adoption remains critically low, creating urgency for early movers.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€4.7bn €602m – €31.0bn	€3.4bn €432m – €22.2bn	€204m €25.9m – €1.3bn	partial
Observed: tendered contract value, annualised	€20.9m €20.9m – €56.6m	€20.9m €20.9m – €56.6m	€1.3m €1.3m – €3.4m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Manufacturing (EU27_2020)	238,122 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Post-quantum cryptography	35.8 % of enterprises (10+ employees)	proxy	Eurostat · isoc_cisce_ran2 · E_SECMDENC	2024
Annual value of one engagement	€337k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Size-weighted buyer base	39,220 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	6.0 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. ICT security measures by NACE Rev. 2 activity series E_SECMDENC is used as a proxy for Post-quantum cryptography: Encryption users are the migration population. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 2 of 3 declared geographies are outside the European reference data (US, IN); the estimate covers the European footprint only.

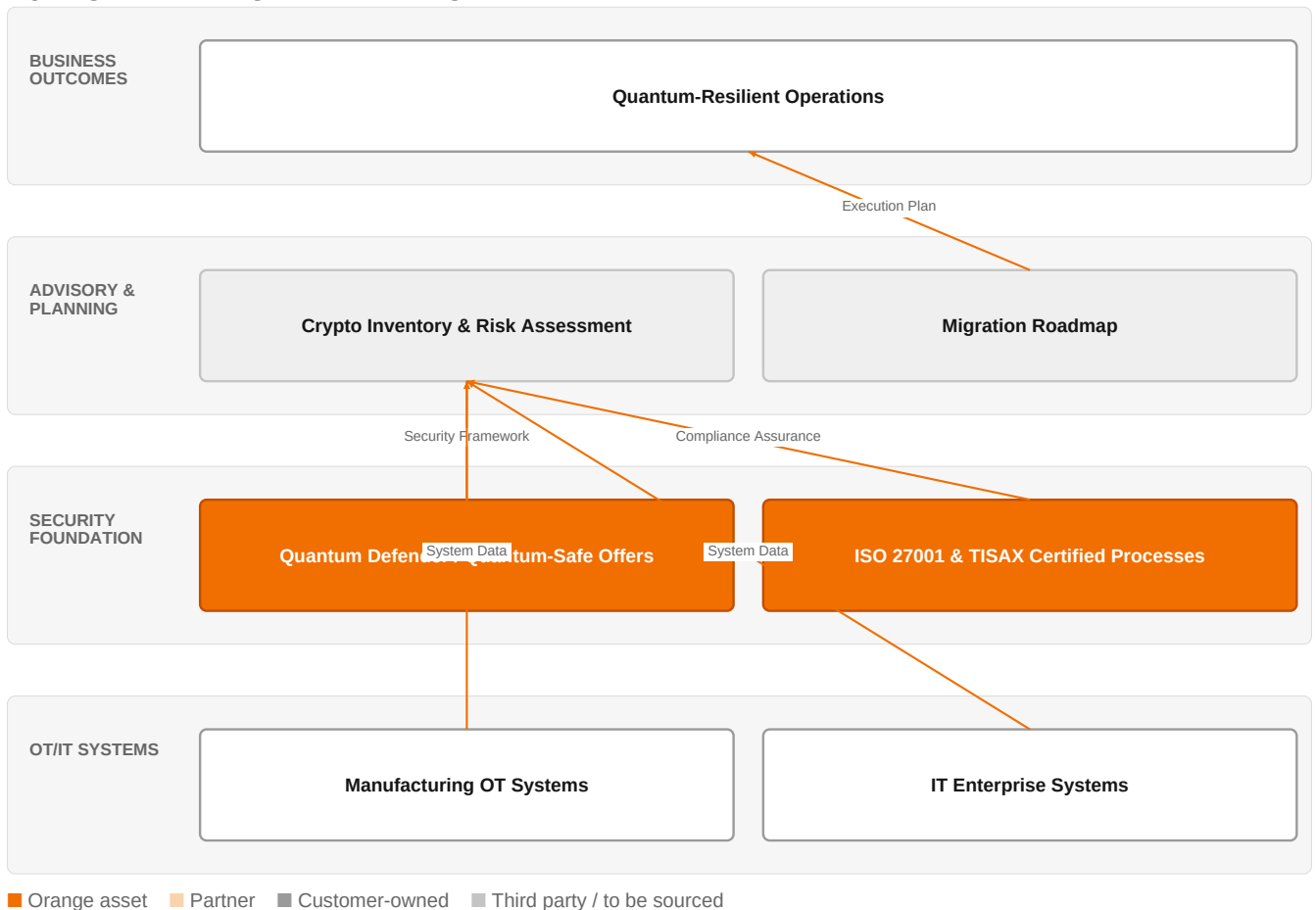
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€20.9m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	6.0 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 14 matching notices, matched on vertical via the CPV crosswalk — \$4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 2 of 3 declared geographies are outside the European reference data (US, IN); the estimate covers the European footprint only.

The solution, in one picture

PQC Migration Planning for Manufacturing



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

Orange's advisory-led approach maps OT/IT systems, applies certified security processes, and delivers a migration roadmap to achieve quantum-resilient operations.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

NIST has finalized PQC standards (FIPS 203, 204, 205), yet enterprise adoption remains critically low, especially in Global Capability Centres that process sensitive cross-border data. Large-scale quantum computers threaten current public-key cryptography, and adversaries can already record encrypted traffic for future decryption. Migration is a multi-year, multi-domain transformation across cloud, enterprise, embedded, and OT systems, requiring unprecedented ecosystem coordination. Governments and standards bodies have published transition roadmaps, but policies diverge, and most enterprises lack operational models to prioritize exposure.

Sources: SIG-B5F1E5C72CAB Springer Science and B 2026-07-30; SIG-B8561992126B openalex.org 2026-07-30; SIG-57F4F0DAE044 Cryptography 2026-06-18; SIG-8BD0B85D3181 openalex.org 2025-12-24; SIG-D45704585F54 Elsevier BV 2026-01-01; SIG-2FE78254EE94 arxiv.org 2026-07-31

Who buys, and why now

The Chief Information Security Officer (CISO) or Head of IT Security signs, but the pain is felt by OT/IT architects and compliance officers who must inventory thousands of applications and cryptographic primitives. The trigger is often a regulatory deadline, a customer audit, or a board-level risk assessment highlighting quantum exposure. They need to move from 'we know we need PQC' to a concrete, auditable migration plan, covering crypto discovery, crypto-agility decisions, hybrid deployment sizing, and PKI impact modeling.

Why it is hot now — every claim bound to a dated source

- Large-scale quantum computers threaten the public-key cryptography that protects enterprise data, communications, and digital identity. [SIG-B8561992126B]

- Enterprises operating large-scale networks must migrate thousands of applications that rely on cryptographic primitives, each requiring distinct remediation strategies and varying effort. [SIG-CE5A95F7AF72]
- The emergence of quantum computing threatens the security of classical cryptographic algorithms such as RSA and ECC. [SIG-8BD0B85D3181]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a team of Cybersecurity experts and Defense and security experts to conduct a crypto inventory and risk assessment, leveraging Orange Group researchers for the latest PQC insights. The engagement would follow a phased approach: first, discovery and inventory of cryptographic assets; second, risk prioritization using a quantum risk index; third, migration planning with hybrid deployment sizing; and finally, execution support with ticket-ready waves. This would be delivered under ISO 27001 and TISAX certifications, ensuring trust and compliance. The Quantum Defender / quantum-safe offers would provide the underlying security framework.

Why Orange can win

Orange brings a unique combination of deep cybersecurity expertise, defense-grade security credentials, and research capabilities from Orange Group researchers. The ISO 27001 and TISAX certifications provide the trust needed for manufacturing clients, and the Quantum Defender / quantum-safe offers are directly relevant. However, the assets are thin on specific PQC migration tools; Orange would need to partner or build on existing capabilities to compete with specialists.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Quantum Defender / quantum-safe offers	offer	L0	offer addresses the use case AND provides the technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
TISAX	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
Cybersecurity experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Colgate-Palmolive	reference	SUP	published reference in this vertical, different use case	unconfirmed
Mondelez International	reference	SUP	published reference in this vertical, different use case	unconfirmed
Saint-Gobain Glass	reference	SUP	published reference in this vertical, different use case	unconfirmed
Skytale	reference	SUP	published reference in this vertical, different use case	unconfirmed

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

Competitive landscape

MEDIUM competitive intensity (score 6.4; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitors include global systems integrators like Atos/Eviden, IBM, Accenture, and Capgemini, who have broad enterprise reach and existing manufacturing relationships. Security specialists like Airbus Protect, Entrust, ID Quantique, and Thales offer deep cryptographic expertise and may have more mature PQC offerings. The customer will compare Orange's end-to-end trust and sovereignty positioning against these specialists' technical depth and the integrators' scale.

Sources: SIG-57F4F0DAE044 Cryptography 2026-06-18; SIG-8BD0B85D3181 openalex.org 2025-12-24

Competitor	Category	Position here	Basis	Evidence
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Atos / Eviden	Global systems integrator	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
IBM	Global systems integrator	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
Airbus Protect	Security specialist	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
Entrust	Security specialist	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
ID Quantique	Security specialist	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
Thales	Security specialist	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
Accenture	Global systems integrator	active in Manufacturing	structural	—
Capgemini	Global systems integrator	active in Manufacturing	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 Have you inventoried the cryptographic algorithms used across your OT and IT systems?
- 2 Do you have long-lived industrial data (e.g., product designs, process data) that must remain secure for decades?
- 3 Have you received any customer or regulatory requests regarding post-quantum readiness?
- 4 What is your current crypto-agility—how quickly can you swap cryptographic algorithms in your OT devices?
- 5 Have you assessed the quantum risk to your supply chain or intellectual property?
- 6 Do you have a migration plan for NIST PQC standards, or are you still in the awareness phase?

Objections you will meet

They will say	You can answer
Quantum computers are years away; why should we invest now?	It's true that a cryptographically relevant quantum computer may be years away, but adversaries can already record encrypted data today to decrypt later. Migration is a multi-year effort, so starting now is prudent.
We already have a cybersecurity team; why do we need Orange?	Your team likely has expertise in classical security, but PQC migration requires specialized knowledge of NIST standards, hybrid schemes, and OT-specific challenges. Orange brings certified experts and a structured methodology.
We're considering a big integrator like IBM or Atos for this.	Those firms have strong offerings, but Orange offers a unique combination of defense-grade security, sovereign infrastructure, and manufacturing vertical expertise, backed by ISO 27001 and TISAX certifications.

Risks and unknowns

The main risk is that PQC migration is a multi-year effort with evolving standards and unclear regulatory timelines, which could stall projects. There is also uncertainty about the actual quantum threat timeline and the maturity of PQC tools for OT environments. Orange's specific PQC migration methodology is not fully defined in the supplied assets, which could be a gap against competitors with proven frameworks.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a manufacturing client, differentiating our approach by combining ISO 27001 and TISAX certifications with the expertise of our cybersecurity and defense experts, and referencing our work with Colgate-Palmolive, Mondelez International, and Saint-Gobain Glass to illustrate our understanding of industrial environments.
Sales	In a conversation with a manufacturing CIO, say: 'As quantum computing advances, the cryptography protecting your long-lived industrial data will need to evolve. Our quantum-safe offers, backed by Orange Group researchers, can help you plan a migration that secures your data for decades.'

Strategist / Innovator	Commission a deep-dive brief on post-quantum cryptography migration for manufacturing OT/IT, leveraging Orange Group researchers to map the current cryptographic inventory and remediation pathways for long-lived industrial data.
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Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-40B64E273856	2026-07-30	International Journal of ...	1	Post-Quantum Cryptography Migration for Enterprise Security
SIG-B5F1E5C72CAB	2026-07-30	Springer Science and Busi...	1	Post-Quantum Cryptography Migration Readiness in Global Capability Centres: A Governance Framew...
SIG-B8561992126B	2026-07-30	openalex.org	1	Post-Quantum Cryptography Migration for Enterprise Security
SIG-57F4F0DAE044	2026-06-18	Cryptography	1	Mythos-Class Frontier Models and the Compression of Post-Quantum Cryptography Migration Timelin...
SIG-C822609B6A07	2026-05-31	openalex.org	1	Large Scale Actualisation of Post-Quantum Cryptography Migration
SIG-BFFA5CE48BF5	2026-02-27	Shanlax International Jou...	1	Migration Strategies from Traditional Cryptography to Post-Quantum Cryptography in Cloud Systems
SIG-964C3B0C145D	2026-02-19	openalex.org	1	Post-Quantum Cryptography - Migration & Crypto Inventory Lab
SIG-CE5A95F7AF72	2026-01-06	openalex.org	1	PQC Planner: Post-Quantum Cryptographic Migration Planner
SIG-353070727CBB	2026-01-01	openalex.org	1	Institutional Approaches to Post-Quantum Cryptography: A Comparative Analysis of Migration Fram...
SIG-D45704585F54	2026-01-01	Elsevier BV	1	Towards a Quantum Risk Index: a risk-based framework for prioritizing Post-Quantum Cryptography...
SIG-D81DF125FECB	2026-01-01	Proceedings of the 23rd I...	1	A Capability-Driven Framework for Prioritizing Post-Quantum Cryptography Migration in Enterpris...
SIG-8BD0B85D3181	2025-12-24	openalex.org	1	Enterprise Migration to Post-Quantum Cryptography: Timeline Analysis and Strategic Frameworks
SIG-ED58CE7F1766	2025-11-24	openalex.org	1	Estimating Migration Timelines for Enterprises Transitioning to Post-Quantum Cryptography: A Re...
SIG-DD1DB909E321	2026-08-03	arxiv.org	3	Measuring Post-Quantum TLS Deployment Across UK Internet Sectors
SIG-2FE78254EE94	2026-07-31	arxiv.org	3	Mind the Gap: Policy vs Reality in Post-Quantum TLS Deployment

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:18+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:40:42+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.